

Batch emissions report

exp1_llm_only_2026-10-09_20-20-05

Models: llama3

Question ordering: Keep original order

Shuffle seed: None

Pipeline: LLM only | Questions: 3 | Successful: 3 | Failed/unknown: 0

Batch runtime: 293.44 s | Successful requests/min: 0.61342

Measured cost (all requests)	Observed sum	Valid / all
Energy (Wh)	2.4104	3 / 3
Emissions (g CO2e)	1.7196	3 / 3

Method and data quality

Statistics and distributions use successful requests with finite measurements. Cost totals include measured failed requests. Missing/non-finite values are excluded, never replaced with zero. Partial totals are not estimates of unmeasured cost.

Variance and standard deviation use the sample definition ($n - 1$); both are N/A for fewer than two observations. Variance has squared units. P95 uses linear interpolation. Small values use significant digits or scientific notation.

Historical pipelines may already have stored missing telemetry as zero. This report cannot distinguish those defaults from measured zero. RAM fields describe estimated energy, not memory usage. Per-question charts use CSV row order; red points mark failed/unknown requests.

Metric	Missing	Zero
Latency (s)	0	0
Energy (Wh)	0	0
Emissions (g CO2e)	0	0
CPU energy (Wh)	0	0
GPU energy (Wh)	0	3
RAM energy (Wh)	0	0

Measurement scope and reliability

Machine scope includes supported CPU/GPU/RAM activity from local Ollama and other processes; it does not isolate model energy or measure wall-socket whole-system power. Unknown backends are unverified; RAM is an energy estimate, not memory usage. A local endpoint does not prove all accelerator hardware is supported.

Tracking sessions recorded: 3. Expected for this CSV: 3. Metadata describes configuration and backend evidence, not independent meter validation.

codecarbon_version: 3.3.0

os: Windows-11-10.0.26200-SP0

effective_scope: machine

measure_power_secs: 1

pue: 1.0

country_iso_code: IND

Config Profile: "inference"

Configured Settings: {"allow_multiple_runs": true, "country_iso_code": "IND", "log_level": "error", "measure_power_secs": 1, "pue": 1.0, "save_to_api": false, "tracking_mode": "machine"}

CPU backend: unknown / not exposed by this version

GPU backend: unknown / not exposed by this version

RAM backend: unknown / not exposed by this version

Hardware: {"cpu_model": "12th Gen Intel(R) Core(TM) i5-1235U", "gpu_model": "", "gpu_count": 0, "ram_total_size": 15.69796371459961}

Backend objects/modes: [{"class": "RAM", "mode": "unknown"}, {"class": "CPU", "mode": "windows_emi"}]

Sensor-backed versus estimated: backend names containing constant, TDP or CPU load indicate estimation; RAPL, NVML and power metrics indicate hardware telemetry. Unknown modes remain unverified. Emissions are calculated estimates in all cases.

Tracker stop failures: 0

A partial observed sum excludes unavailable readings. A complete batch cost cannot be claimed when coverage is incomplete. Zero readings are retained; unknown hardware coverage must not be interpreted as zero consumption.

Statistical summary

Latency (s)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
3	97.682	93.394	1328.2	36.445	63.572	136.08	131.81

Energy (Wh)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
3	0.80345	0.76775	0.09002	0.30003	0.52286	1.1197	1.0845

Emissions (g CO2e)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
3	0.57321	0.54775	0.04582	0.21406	0.37303	0.79887	0.77375

CPU energy (Wh)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
3	0.53943	0.51271	0.038488	0.19618	0.35798	0.7476	0.72411

GPU energy (Wh)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
3	0	0	0	0	0	0	0

RAM energy (Wh)

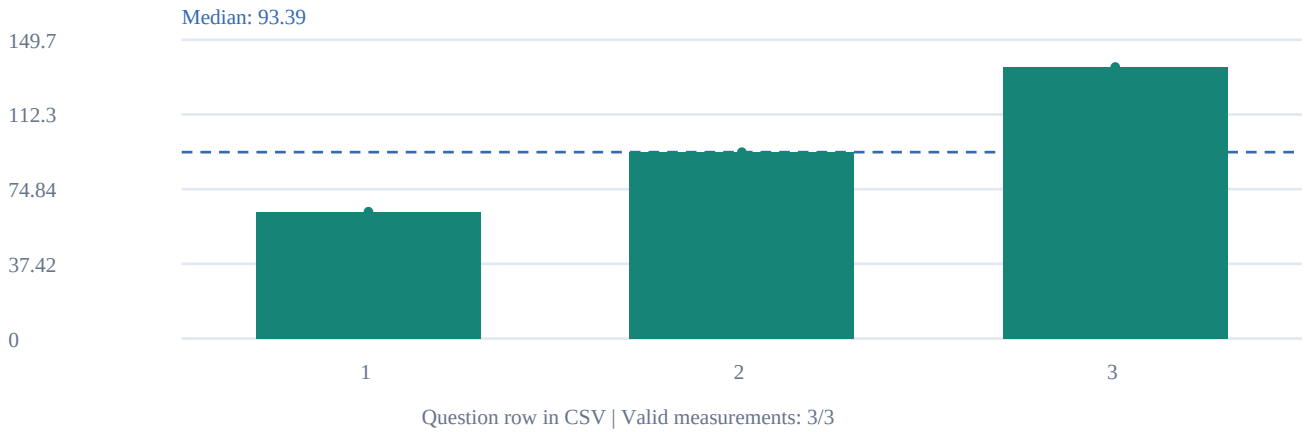
Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
3	0.26402	0.25505	0.010798	0.10391	0.16489	0.37213	0.36042

Which questions cost the most?

Bars show individual requests; large batches use points to avoid overcrowding. Missing measurements are marked X, not shown as zero.

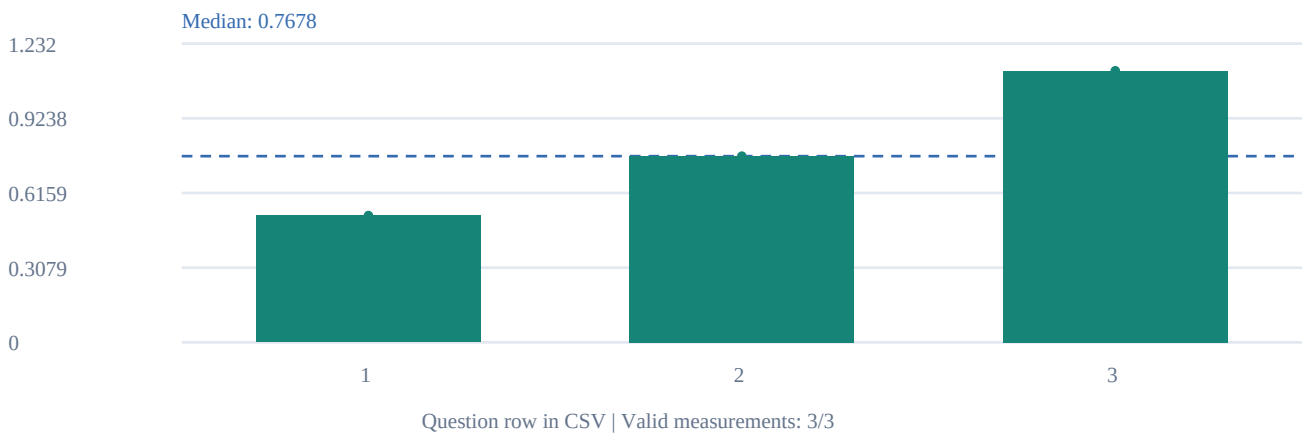
Latency (s)

Teal: successful | Red: failed/unknown | X: missing | Blue line: successful median



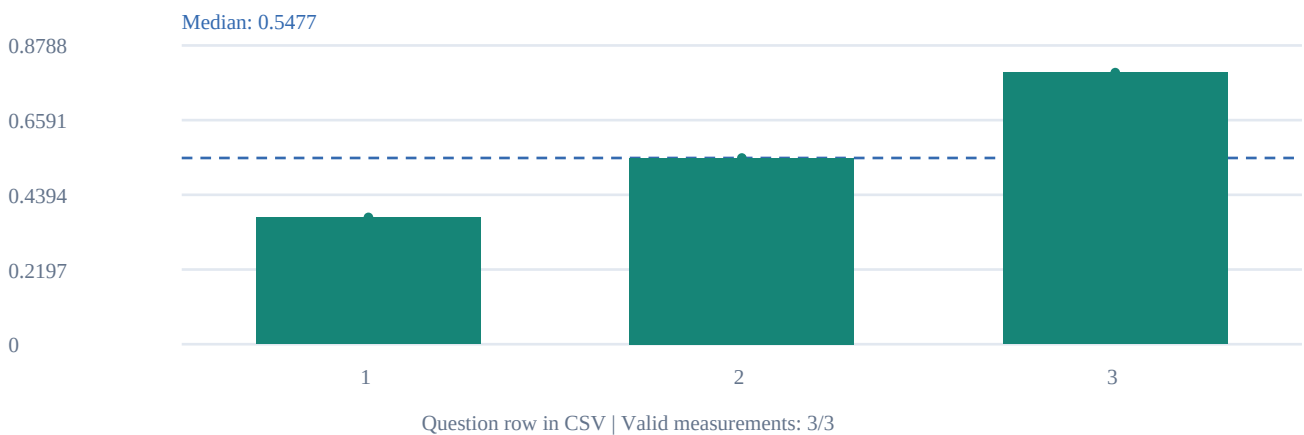
Energy (Wh)

Teal: successful | Red: failed/unknown | X: missing | Blue line: successful median



Emissions (g CO2e)

Teal: successful | Red: failed/unknown | X: missing | Blue line: successful median

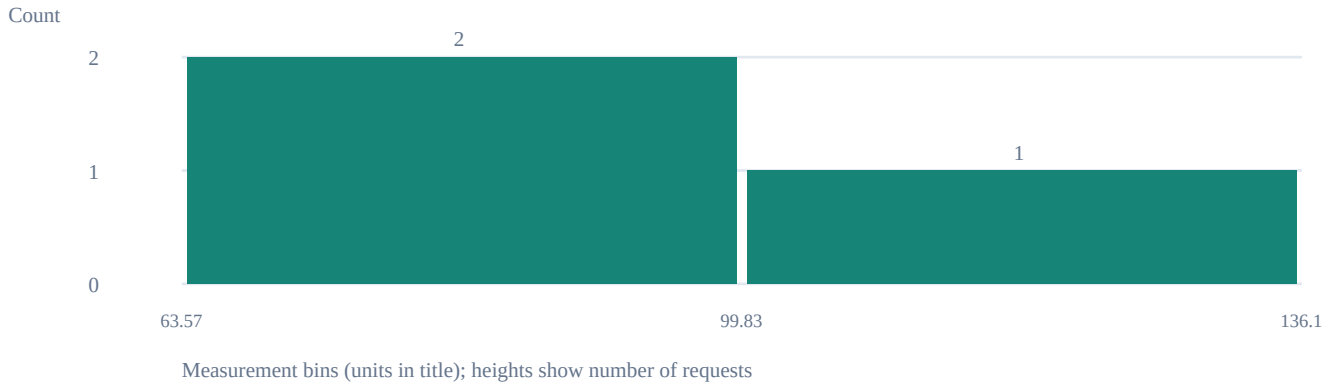


How much do requests vary?

Successful requests only. Bin boundaries show measurement values; bar heights and labels show request counts.

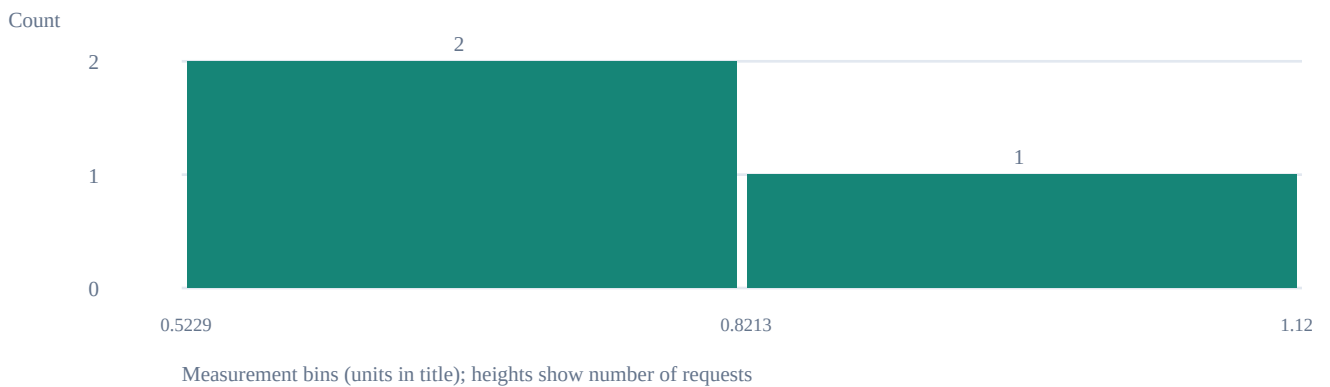
Latency (s)

Successful requests: n=3 | Mean 97.68 | Median 93.39



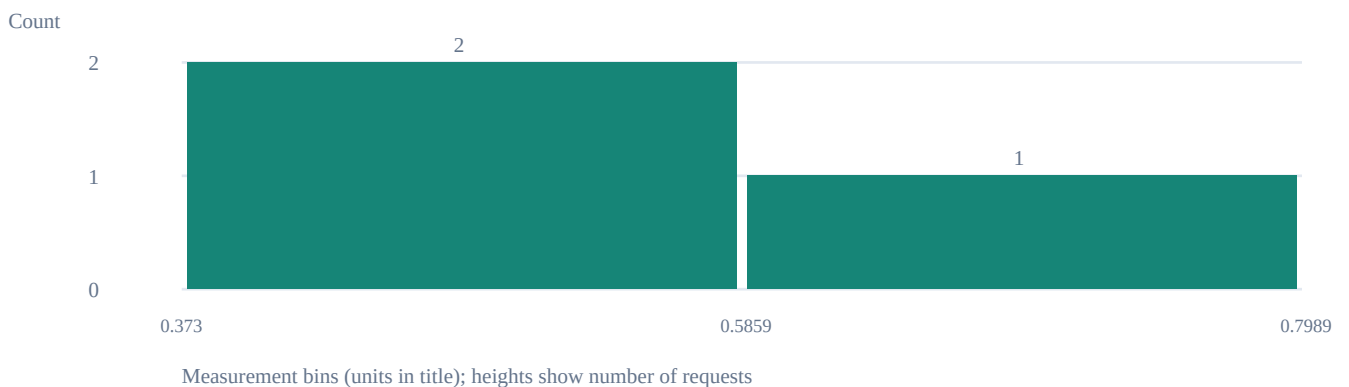
Energy (Wh)

Successful requests: n=3 | Mean 0.8035 | Median 0.7678



Emissions (g CO2e)

Successful requests: n=3 | Mean 0.5732 | Median 0.5477



Where is energy spent?

Compare observed component costs below. Amber bars indicate incomplete coverage; comparisons with different coverage may be misleading. These are tracking estimates, not isolated model or wall-socket measurements.

Hardware energy (Wh)

Observed sums across all requests; partial coverage is labelled.

CPU		1.618
GPU	Valid 3/3	0
RAM	Valid 3/3	0.7921

Question execution mapping

Chart question rows correspond to this execution sequence. Original serial numbers are local to each topic sheet.

Position	Topic sheet	Serial	Question type
1	General	7	Closed-Book Scenario
2	General	8	Closed-Book Scenario
3	General	9	Closed-Book Scenario